

ASTE-331: Spacecraft Systems Engineering

Thermal

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• Function

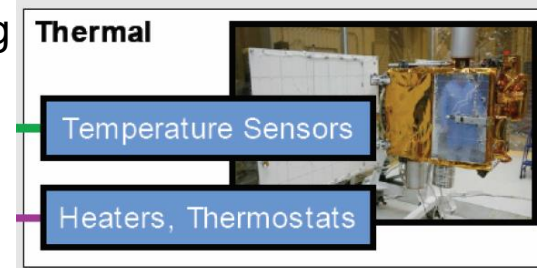
- Provide appropriate thermal control throughout the spacecraft
- Primary driver is the overall thermal model that includes temperatures from:
 - Avionics (usually room temperature)
 - Instruments (eg, cold FPAs or hot sample analyses)
 - Batteries, mechanisms, propulsion
- All components are required to stay within their qualification temperature limits

• Common Components

- Surface coatings & treatments
 - Films, paints, MLI
- Conductors and insulators
- Heat transfer regulators
 - Heat pipes, louvers, phase-change devices, heat switches

• GRAIL Example

- Paint, films
- Multi-layer insulation (MLI)
- Instrument thermal mass plate
- Heaters, thermostats, sensors, etc.
- Total = ~ 10 kg



• Key Trades & Analyses

- Active versus passive
- Thermal analysis
- Radiator design, locations, & margins

• Key Parameters

- Mass, power, & cost
- Updated attitude profile

Thermal Control System Scope

- **Maintain spacecraft system elements within specified temperature limits**
- **Levels limited to support**
 - Chemistry (batteries, thrusters)
 - Phase/viscosity (propellants, lubricants)
 - Thermal stress (electronics, alignments)
 - Mechanical properties
 - Other (low temperature sensors)

Architecture Objectives

- **Maximize System Operability**
 - **Pointing:** allow spacecraft to freely point without exceeding thermal limits
 - **Power:** Minimize impact of constraints on operability (eg, relationship with battery & array temperature)
- **Provide Robustness**
 - Insensitive to environment and system characteristics
 - Insensitive to small design changes
 - Simplifies testing...



Typical Subsystem Requirements

Updated Spacecraft Component Temperature Table

Component / Subsystem	Operating Temperature (°C)	Survival Temperature (°C)
Digital Electronics (modern avionics, rad-hard CPUs, FPGAs)	-20 to +70	-40 to +85
Analog Electronics	-20 to +55	-40 to +85
Batteries (modern Li-ion spacecraft cells)	0 to +30	-20 to +50
IR Detectors (HgCdTe, bolometers, cooled arrays)	-253 to -173 (20-100 K)	-253 to +30
Solid-State Particle Detectors (Si PIN, CdTe)	-40 to +20	-50 to +40
Momentum Wheels / Reaction Wheels / Motors	-20 to +60	-30 to +70
Solar Panels (modern triple-junction GaAs)	-150 to +135	-180 to +150

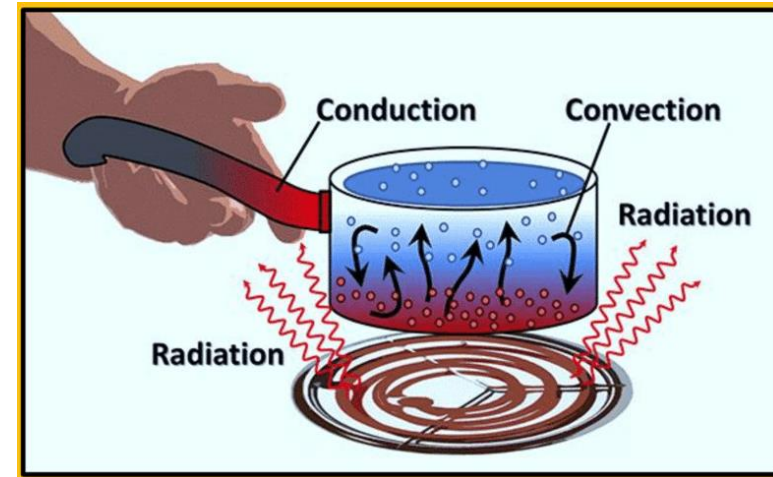
Thermal Control Environments

Updated Mission Phase Environment Table

Phase	Time Period	Environment
Ground Operations	Months to years before launch	Cleanroom (ISO-8 to ISO-5), 18–23 °C, controlled humidity (30–50%), transportation vibration/shock, EMI/EMC test environments, thermal-vac testing, integration & test (I&T) facilities
Launch	~10–15 minutes ascent + ~5–20 minutes coast	High acoustic loads (140–150 dB), intense vibration, pyro-shocks, fairing thermal gradients, plume heating, rapidly changing g-loads (0–6 g), RF environment from launch vehicle
Transfer Orbit	Hours to weeks (LEO → GEO, lunar, planetary, cruise)	Sun/Earth/Moon thermal cycles, albedo variations, eclipse periods, outgassing and contamination, arbitrary or constrained attitudes, initial propulsion burns, higher radiation (Van Allen belts for GEO transfers)
Final Orbit / Operational Phase	Years to decades	Stable or cycling thermal environment depending on orbit, solar cycles, radiation (SEEs, TID), seasonal beta angles, varying attitudes, payload-driven modes, station-keeping, end-of-life degradations (solar array, batteries, lubricants)

Thermal Design Practices

- **Design approaches are based on:**
 - **Conduction** typically dominates at the component level
 - **Radiation** dominates at the system level
 - **Convection** is generally not applicable
 - Due to vacuum / zero-g environment



- **Good Practices**
 - Co-locate components/systems with similar requirements
 - Place-high-power devices near exterior (radiators)
 - Point radiators towards deep space
 - Limit view factors of radiators
 - View factors are the fraction of other spacecraft structure/components in FOV
 - Cold-biasing: design regions of spacecraft to operate below design temperature and use heaters to provide positive control (provides margin)

Conservation of Energy

- **Reflectivity (ρ)**

- The fraction of incident radiant energy that is reflected by a surface.
- Range: 0 to 1
- $\rho = 1$ means all incoming energy is reflected
- $\rho = 0$ means no energy is reflected

- **Absorptivity (α)**

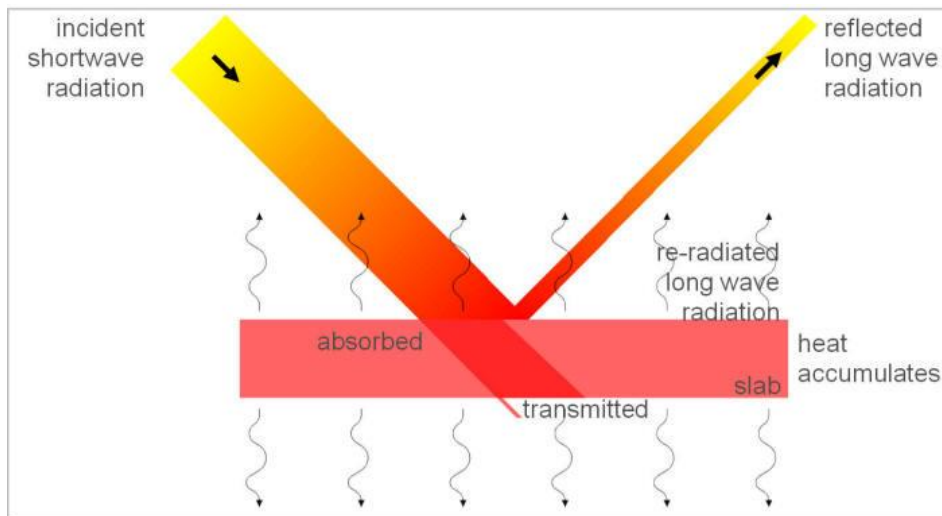
- The fraction of incident radiant energy that is absorbed by a surface.
- Range: 0 to 1
- $\alpha = 1$ means all incoming energy is absorbed
- $\alpha = 0$ means no energy is absorbed

- **Transmissivity (τ)**

- The fraction of incident radiant energy that passes through a material.
- Range: 0 to 1
- $\tau = 1$ means *all* the energy is transmitted (passes through)
- $\tau = 0$ means *none* is transmitted

- **Emissivity (ϵ)**

- A measure of how effectively a surface emits thermal radiation compared to an ideal blackbody.
- Range: 0 to 1
- $\epsilon = 1$ means the surface behaves like a perfect blackbody (maximum emission/absorption)
- For a given wavelength and direction:
- $\alpha = \epsilon$ (Kirchhoff's Law of Thermal Radiation)



Conservation of Energy

$$\text{Reflectivity} + \text{Absorptivity } (\alpha) + \text{Transmissivity } (\tau) = 1$$

Note that transmissivity is typically assumed to be 0.

Surface Coatings & Treatments

- **Because spacecraft thermal balance in space is dominated by radiation, the optical properties of external surfaces are critical drivers of temperature control.**
- **To manage absorptivity, emissivity, and reflectivity, spacecraft use a variety of surface coatings and treatments, including:**
 - Thermal control films and paints (e.g., white paints, OSRs, Kapton)
 - Second-surface mirrors (high reflectivity, low absorptivity)
 - Multi-Layer Insulation (MLI) for minimizing radiative heat transfer
 - Anodized surfaces and other chemical surface treatments

Table A1: Material Properties for Common S/C Surfaces

Surface Properties	Absorptivity	Emissivity	Comments
	α	ϵ	
Optical Solar Reflector	0.07	0.8	8 mil Quartz mirrors
White Paint	0.22	0.85	S13G-LO
Black Paint	0.97	0.84	3M Black Velvet
Aluminized Kapton	0.38	0.67	1 mil
Metallic	0.13	0.04	Vapor Deposited Aluminum
MLI (white beta cloth cover)	0.45	0.04	Estimate (online source), $\epsilon = \epsilon^*$
MLI (aluminized beta cloth cover)	0.37	0.04	" "
MLI (tedlar reinforced cover)	0.3	0.04	" "
MLI (teflon-backed cover)	0.1	0.04	" "

Energy Conservation / Thermal Balance

$Q_{out} = Q_{in}$, where

$$Q_{out} = Q_{rad} + Q_{vent}$$

Q_{out} = Total heat leaving the s/c

Q_{rad} = Emitted radiation from the s/c

Q_{vent} = Heat contained within matter that is vented/ejected (typically 0)

$$Q_{in} = Q_s + Q_{rs} + Q_p + Q_d + Q_o$$

Q_{in} = Total heat entering the s/c

Q_s = Direct input from the sun

Q_{rs} = Reflected solar flux (eg, sun → Earth → s/c)

Q_p = Infrared energy from planets (eg, Earth)

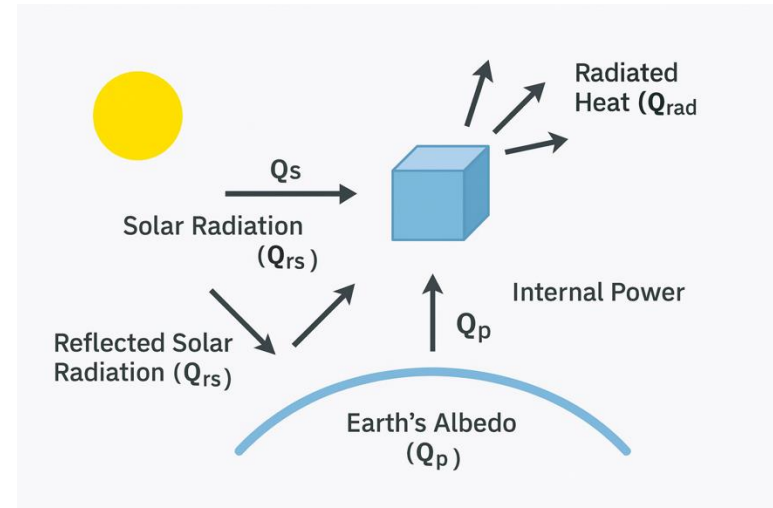
Q_d = Electric power dissipated within the s/c

Q_o = Input power from other elements of the system (eg, arrays); assume 0 for this HW

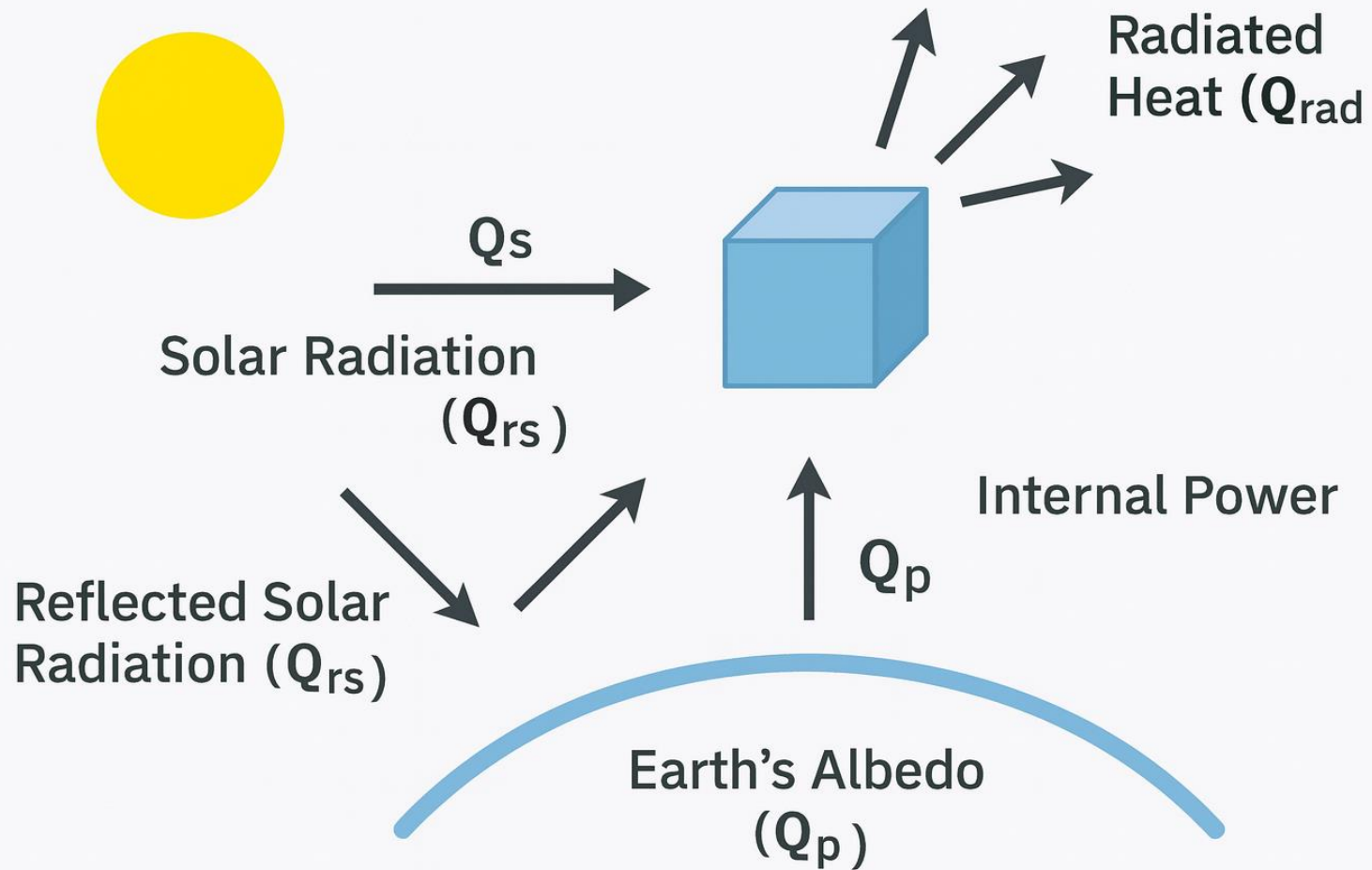
Note that more generically, 'q' is the heat flux (or heat load) across a boundary and is typically measured in W/m^2 , whereas Q is the total heat/power (measured in W) that is transferred.

Thus,

$$Q = A q, \text{ where } A = \text{surface area}$$

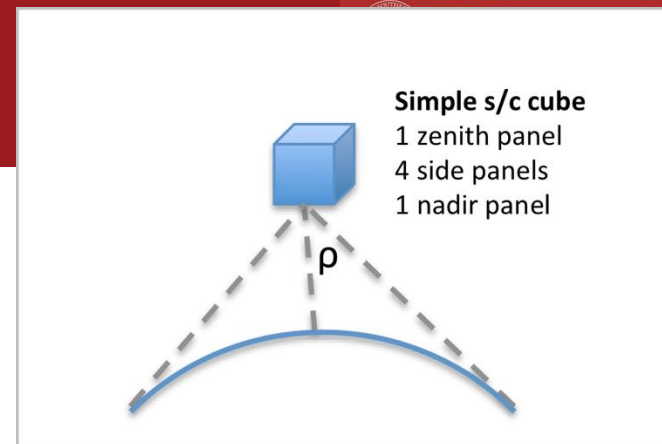


Thermal Power Balance



Radiated Emissions & Solar Flux

Eqn-2	Stefan-Boltzmann Law (radiated emission) $Q_{rad} = A \epsilon \sigma T^4$, where Q_{rad} = radiated heat (W) A = surface area (m^2) ϵ = emissivity of the material (perfect blackbody = 1) σ = Stefan-Boltzman constant ($W / m^2 K^4$) T = temperature of the object (K)
Eqn-3	Direct Solar Flux $Q_s = A G_s \alpha$, where G_s = Solar flux (assume $1,370 W / m^2$ at 1 AU) α = surface absorptivity (unitless)
Eqn-4	Reflected Solar Flux $Q_{rs} = A G_{rs} \alpha$, where G_{rs} = Reflected solar flux (W/m^2) $G_{rs} = a K G_s \sin^2 \rho$, where a = albedo (assume 0.3 for Earth, although this varies; eg, cloud cover) K = Correction factor for reflected Earth energy ($K = 0.665 + 0.521\rho - 0.203\rho^2$) $\sin^2 \rho$ = represents the view factor, ranging from 0 to 1, for how much available energy reaches the s/c $\rho = \arcsin(R_p / r)$, where ρ = angular radius of planet (rad) R_p = radius of planet (km), assume 6,370 km for Earth r = orbit radius (km)
Eqn-5	Infrared Energy from Earth $Q_p = A G_p \alpha$, where $G_p = \epsilon_p \sigma T_p^4 \sin^2 \rho$, where ϵ_p = emissivity of planet (eg, Earth, assume 0.5) T_p = temperature of planet (eg, Earth, assume 300K) α = surface absorptivity (at Earth's IR wavelength, this can be assumed to be same as emissivity; ie, $\epsilon_s = \alpha$)



Effective Area for Planetary View Factor

When a spacecraft is orbiting a planet, the altitude of the s/c will determine the planetary view factor. That is, at a low altitude, the planet will fill much of the 180-deg field of view, whereas at a distance orbit, the planet will only be a small fraction of the field of view. This view factor is measured via the angular radius of the planet (ρ), calculated as part of Eqn-4.

This relationship also leads to an 'effective area' (A_e) depending on the position of the spacecraft surfaces, which may face towards the Earth (nadir), away from the Earth (zenith), or normal to the Earth (side).

Nadir: $A_e = A$, as the surface area receives all of the energy from the Earth (either IR or reflected light)

Zenith: $A_e = A \times 0$, as the surface area does not receive any energy

Side: $A_e = 0.5 A (1 - \cos \rho)$, where 0.5 represents that only half the planet is in view and $(1 - \cos \rho)$ represents the 90-deg rotation of ρ .

Thus, depending on how each surface is orientated, it will have a different effective area for each surface.

Dissipated Power within the S/C

Q_d = Total power dissipated onboard the s/c

Note that this includes heater power and excludes Telecom RF power (which is radiated from the s/c). With respect to RF power, note that the difference between the DC input power & RF power is primarily the power lost to the amplifier inefficiency, which is converted to heat.

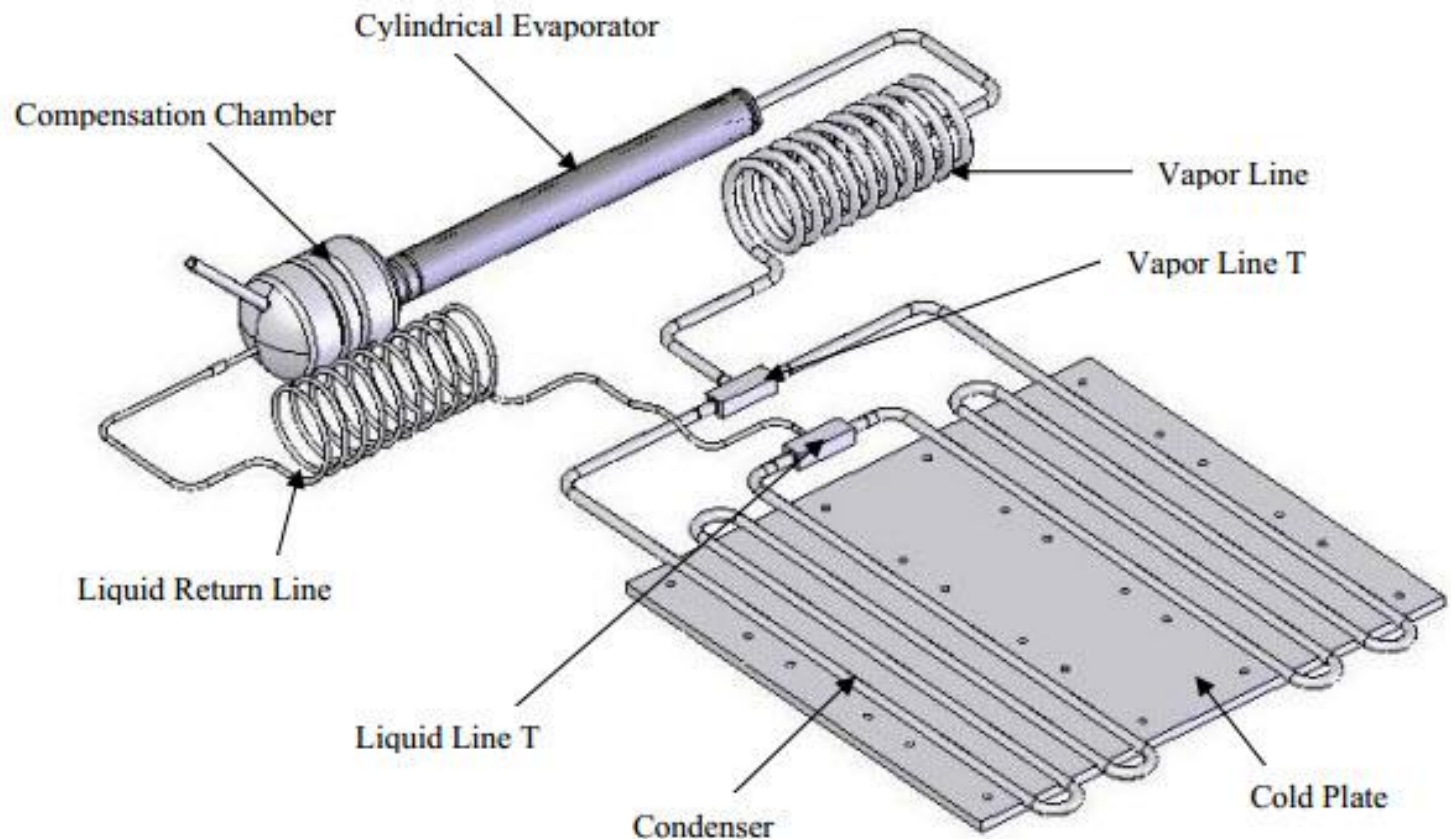
A simplified model of the s/c may be assumed that neglects appendages such as arrays, deployed instruments, antennas, etc. For example, a solar array will deliver a total power to the spacecraft that is primarily dissipated within the electronics. Here, you are just using the power dissipated and NOT (a) any power/heat left on the array or (b) the thermal properties/temperatures of the array itself. Similarly, if there is an instrument on the end of a long boom, you would subtract the power provided this instrument (across the boom/harness interface) from your dissipated power.

Multi-Layer Insulation (MLI)

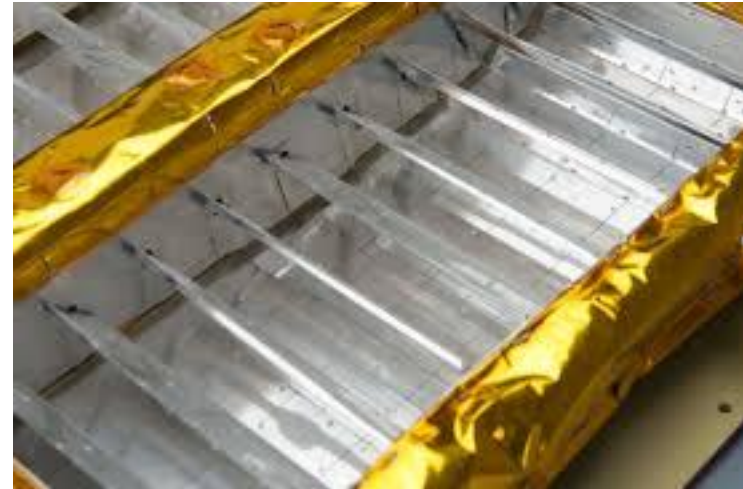
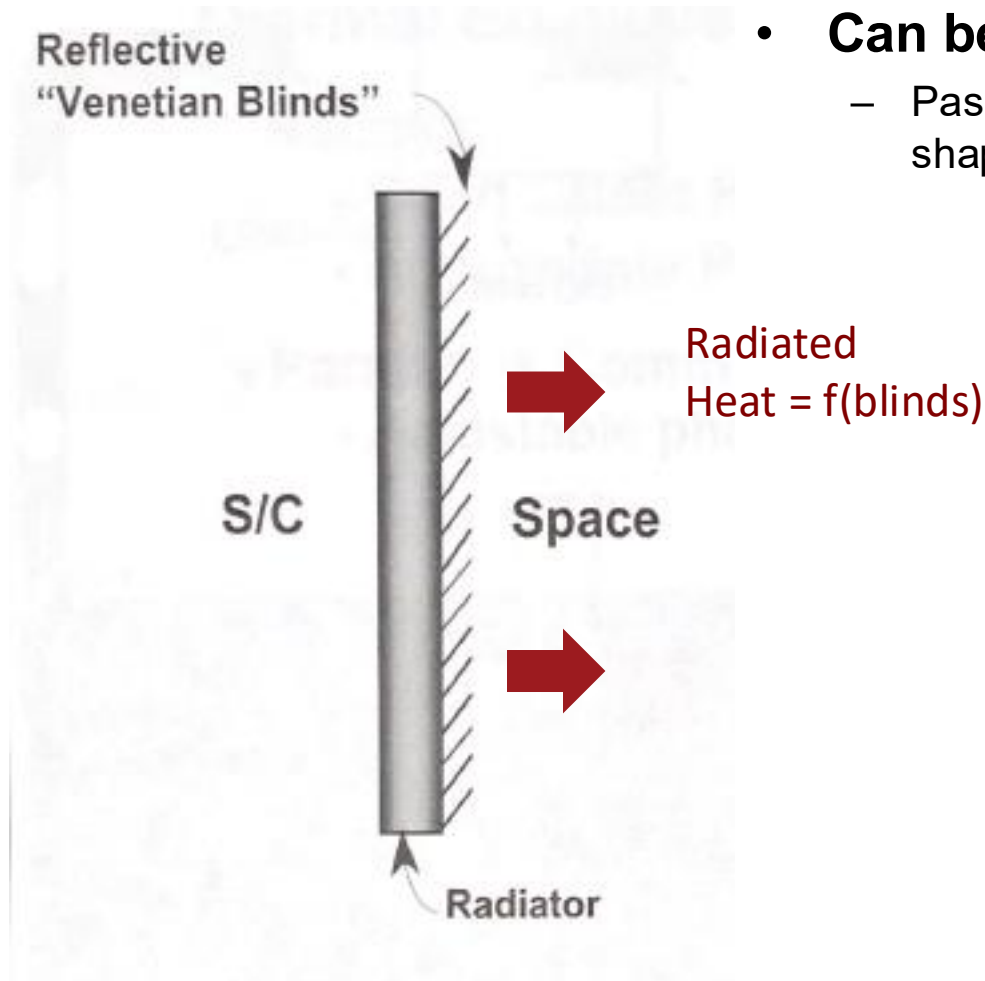


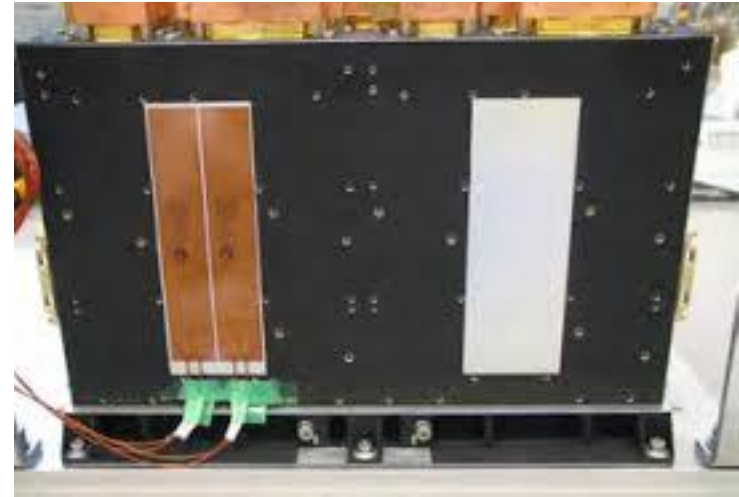
Heat Pipes

- Constant conduction heat pipes transfer heat in one direction
- Loop heat pipes (as shown below) recirculate liquid/vapor

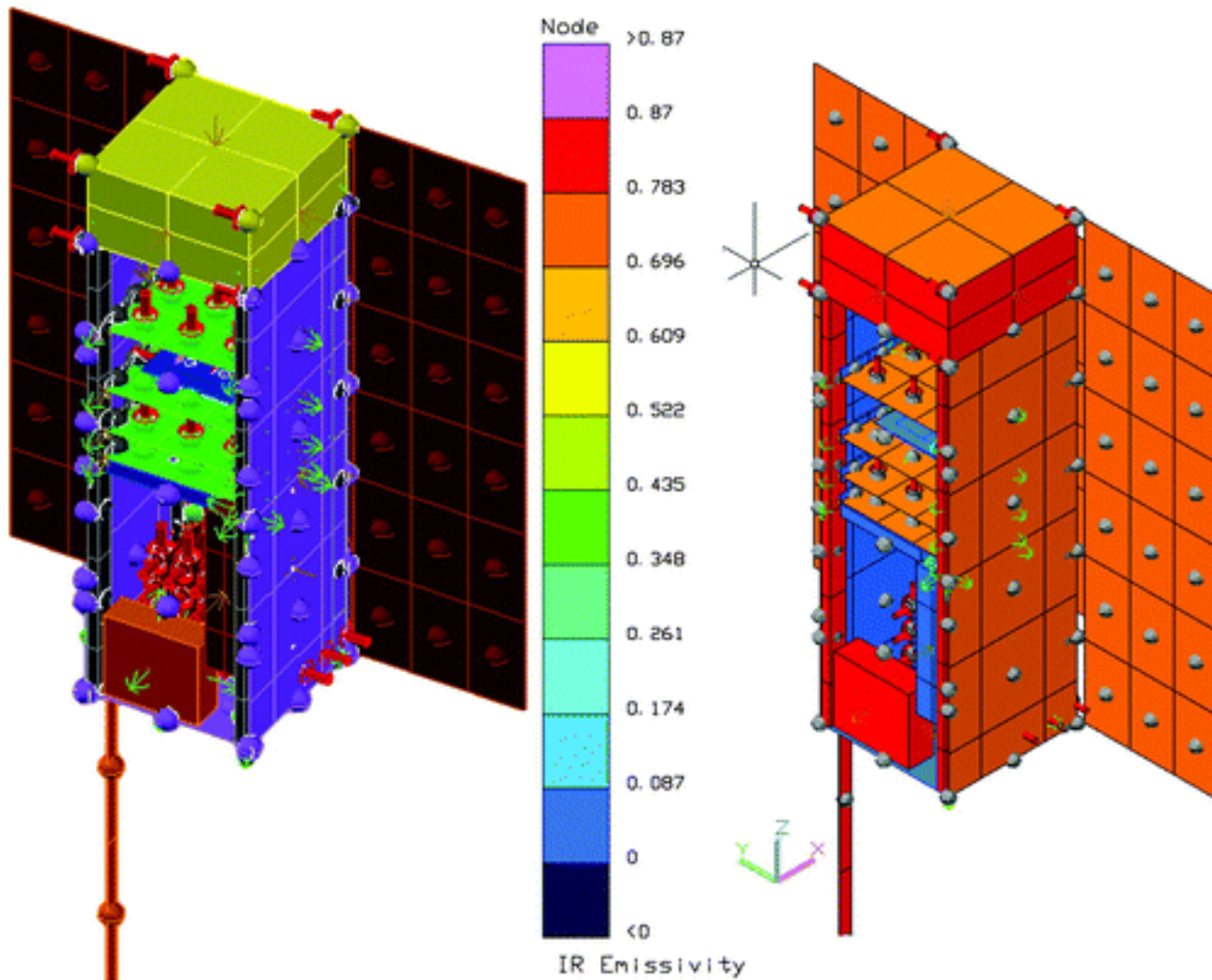


- Can provide 5x reduction in heat transfer
- Can be active or passive
 - Passive approach uses bimetallic strip or shape-memory materials

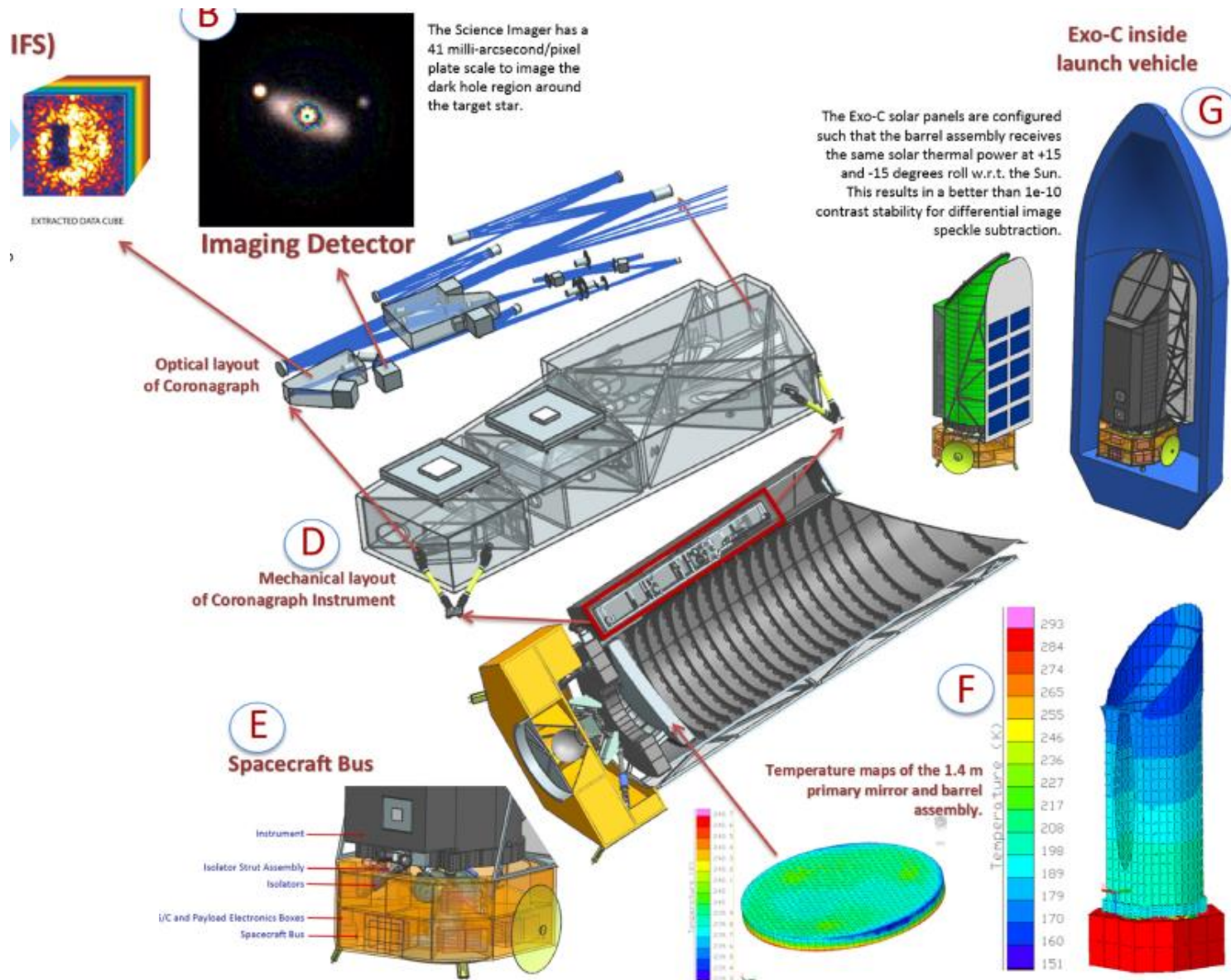




Survival heaters often use mechanical thermostats for control to ensure that independently of flight software, the minimum temperatures are maintained.



Mission Example



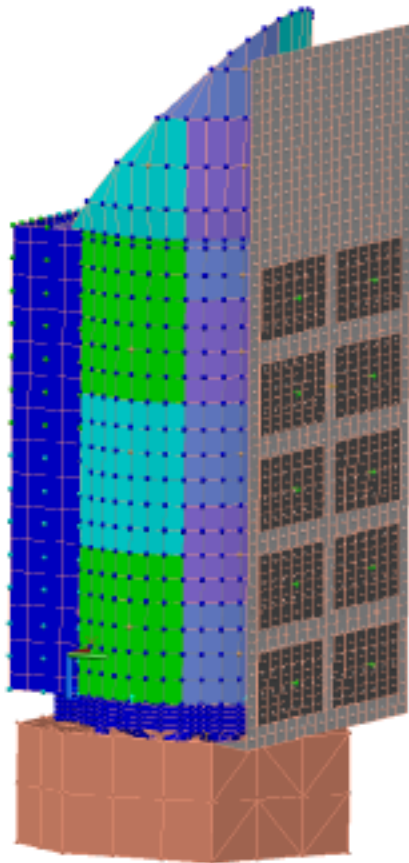
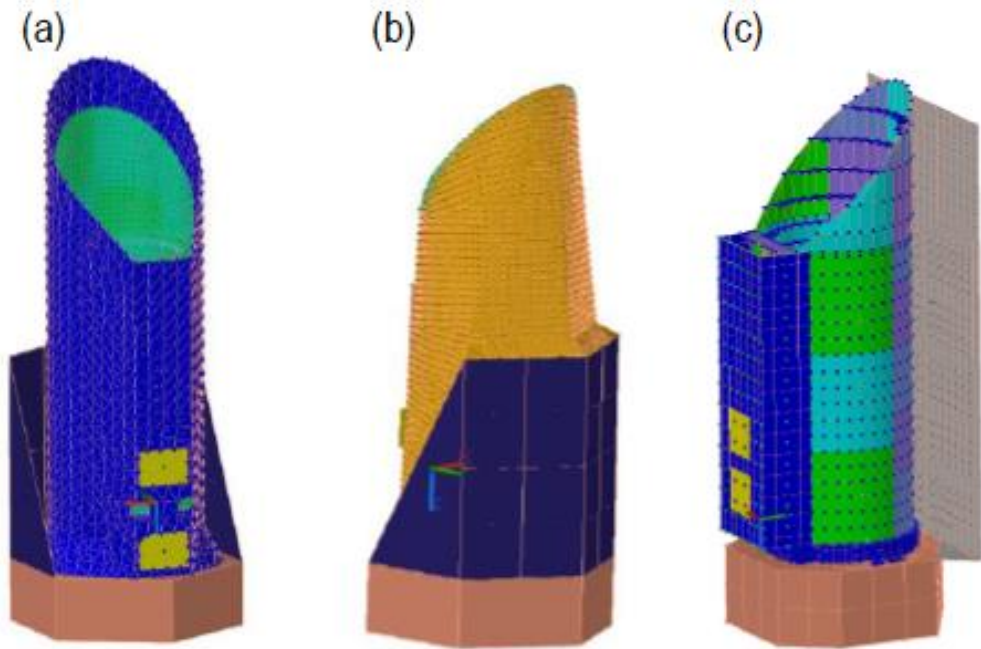


Figure 5.6-1. The Exo-C observatory thermal model. The model includes the spacecraft bus, telescope (including the vibration isolation system), an instrument bench, and a flat solar panel that also serves as a sun-shield. The spacecraft and instrument bench are modeled as constant temperature boundary conditions, and the stability of the telescope was simulated over varying sun angles.

Table 5.6-1. Telescope heater power analysis. The required heater power depends strongly on the set-point temperatures.

Telescope Heater Locations	# of Heaters	Set point (K)	Peak Power (W)
Primary Support Structure	3	200	118
Lower Barrel	18	190	63
Upper Barrel	11	170	56
Barrel Scarf	8	150	111
Primary Mirror	6	240	33
Primary Mirror Bipods	6	240	5
Secondary Mirror	2	240	1
Total	60	-	387
Total with 30% cont.			503

Example Trade Study



Solar Shielding Trade Study	Outer Barrel	MLI-tent	Flat solar panel
Mass	✗	✓	✓
Fast recovery from thermal transient	✗	✓	✓
Contrast stability at steady state	✓	✓	✓
Contrast stability over 30 degree roll and 45 degree pitch	Not analyzed	~2-5 e-9 stability	>1e-10 stability

Figure B.1-29. Three different sunshield configurations: (a) passive outer barrel, (b) Kepler like MLI-tent, and (c) flat solar panel.

Testing in Thermal-Vacuum (TVAC)



M2020 (Perseverance) Rover in JPL TVAC

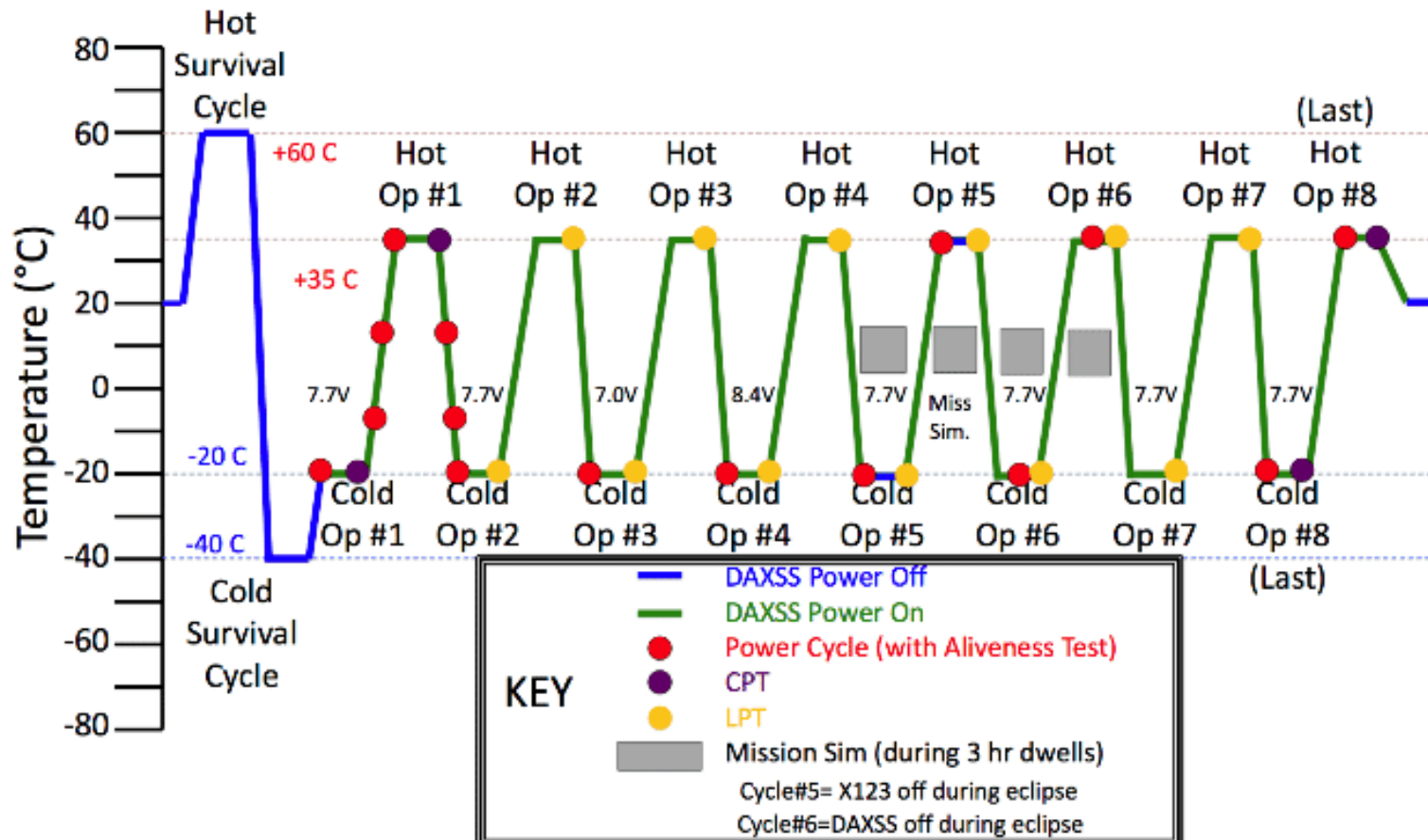
ESA's Euclid Mission



Testing in Thermal-Vacuum (TVAC)

Example Test Plot

DAXSS CubeSat Thermal Vacuum Cycle Test Profile



Thermal Design Steps

- **Review & Understand Design Information**
 - Mission Description and Concept of Operations
 - System and Subsystem Requirements
 - Mission geometry; Payload & subsystem temperature requirements
- **Create a Preliminary Design**
 - Identify a preliminary architecture
 - Select target temperatures (eg, +20/-20 degC, hot/cold bias?)
 - Consider cryogenic elements (3-120K) & operability
 - Consider materials, coatings, radiators, heat pipes
 - Create back-of-the-envelope heat transfer estimates
 - Preliminary sizing of radiators, MLI, heat pipe, etc.
 - Create thermal balance model
 - Optimize with respect to robustness, fault scenarios, etc.
 - Create component mass list
- **Review & Iterate (w/broader team)**
 - Revisit other options (eg, heat straps, heat switches, louvers, etc.)

